

Fundamentals of Image Processing

► Lecture 3: Fourier transform ◀

Master of Computer Science

Sorbonne University

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Context and goals

- Fourier transform (FT): fundamental tool for image processing
- Fourier transform and its applications are discussed in the four next lectures:
 - Today: definition and application
 - Lecture 4: sampling and discrete Fourier transform (DFT)
 - Lecture 5: spatial filtering $\Rightarrow$ filtering in Frequency domain
 - Lecture 6: edge detection and spatial filtering
- Understand the Fourier representation space
 - several representations of a same function: of time (or space) $\Rightarrow$ of frequency
- Definition of mathematical tools (convolution, Dirac, usual FT) crucial for this lecture
- Definition and properties of the 2D Fourier transform
 - Calculus in Fourier spaces, visualize and interpret a spectrum

Context: analysis of 1D signals in the Frequency domain

Fourier Series

Fourier transform of signals (1D)

Fourier transform of images (2D)

2D Fourier transform: application to image processing

Decomposition of periodic functions into Fourier series

- Let x be a real (or complex) T -periodic function. We have for $t \in \mathbb{R}$:

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \left(a_k \cos\left(\frac{2k\pi t}{T}\right) + b_k \sin\left(\frac{2k\pi t}{T}\right) \right) \quad (1)$$

- Coefficients a_k et b_k are given by:

$$\begin{aligned} a_k &= \frac{2}{T} \int_0^T x(t) \cos\left(\frac{2k\pi t}{T}\right) dt \\ b_k &= \frac{2}{T} \int_0^T x(t) \sin\left(\frac{2k\pi t}{T}\right) dt \end{aligned} \quad (2)$$

Fourier series: decomposition of periodic functions

Writing x with amplitude and phase

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \left(a_k \cos\left(\frac{2k\pi t}{T}\right) + b_k \sin\left(\frac{2k\pi t}{T}\right) \right)$$

- Amplitude: $r_k = \sqrt{a_k^2 + b_k^2}$
- Phase: θ_k , such that $\cos \theta_k = \frac{a_k}{r_k}$ and $\sin \theta_k = \frac{b_k}{r_k}$
- Equation (1) becomes:

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} r_k \cos\left(\frac{2k\pi t}{T} - \theta_k\right) \quad (3)$$

(remember that $\cos a \cos b + \sin a \sin b = \cos(a - b)$)

Fourier series decomposition of periodic functions

Writing x with the complex Exponential function

Some fundamental recalls:

- Euler formula:

$$\exp(iz) = \cos(z) + i \sin(z)$$

- de Moivre formula:

$$\exp(ikz) = (\cos(z) + i \sin(z))^k = \cos(kz) + i \sin(kz)$$

$\Rightarrow$

$$\begin{aligned}\cos(kz) &= \frac{\exp(ikz) + \exp(-ikz)}{2} \\ \sin(kz) &= \frac{\exp(ikz) - \exp(-ikz)}{2i}\end{aligned}$$

Fourier series decomposition of periodic functions

Writing x with the complex Exponential function (cont'd)

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \left(a_k \cos\left(\frac{2k\pi t}{T}\right) + b_k \sin\left(\frac{2k\pi t}{T}\right) \right)$$

- Using Euler and de Moivre formulas, we derive:

$$\begin{aligned} x(t) &= \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \frac{a_k}{2} \left[\exp\left(\frac{i2\pi kt}{T}\right) + \exp\left(\frac{-i2\pi kt}{T}\right) \right] + \\ &\quad \sum_{k=1}^{+\infty} \frac{b_k}{2i} \left[\exp\left(\frac{i2\pi kt}{T}\right) - \exp\left(\frac{-i2\pi kt}{T}\right) \right] \\ x(t) &= \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2} \right] + \\ &\quad \sum_{k=1}^{+\infty} \exp\left(\frac{-i2\pi kt}{T}\right) \left[\frac{a_k}{2} + i\frac{b_k}{2} \right] \end{aligned}$$

Fourier series decomposition of periodic functions

Writing x with the complex Exponential function (cont'd)

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2}\right] + \sum_{k=1}^{+\infty} \exp\left(\frac{-i2\pi kt}{T}\right) \left[\frac{a_k}{2} + i\frac{b_k}{2}\right]$$

- change of summation index $k' \leftarrow -k$ (2^{nd} sum), $a_{-k} = a_k$, $b_{-k} = -b_k$:

$$\begin{aligned} x(t) = & \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2}\right] + \\ & \sum_{k=-\infty}^{-1} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2}\right] \end{aligned}$$

Fourier series decomposition of periodic functions

Writing x with the complex Exponential function

$$x(t) = \frac{1}{2}a_0 + \sum_{k=1}^{+\infty} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2}\right] + \sum_{k=-\infty}^{-1} \exp\left(\frac{i2\pi kt}{T}\right) \left[\frac{a_k}{2} - i\frac{b_k}{2}\right]$$

Let us define: $c_k = \frac{a_k}{2} - i\frac{b_k}{2}$ and $c_0 = \frac{a_0}{2}$, we have:

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k \exp\left(\frac{2ik\pi t}{T}\right) \quad (4)$$

with:

$$c_k = \frac{1}{T} \int_0^T x(t) \exp\left(\frac{-2i\pi kt}{T}\right) dt \quad (5)$$

- x writes as a linear combination of elementary functions Φ_k :
 - $\Phi_k(t) = \exp\left(\frac{2ik\pi t}{T}\right)$
 - $\frac{k}{T}$ is a “pure” frequency ($k \in \mathbb{Z}$, may be negative)

Vector spaces, function spaces and scalar product

- Let us consider the space of T -periodic functions. Admitted: it is a vector space of infinite dimension (basis size is infinite)
- Projection of a function f on a function g : defined by the scalar (or dot) product, $\langle \cdot, \cdot \rangle$, as:

$$\langle f, g \rangle = \frac{1}{T} \int_0^T f(t) \bar{g}(t) dt \quad \text{with} \quad f, g : \mathbb{R} \mapsto \mathbb{C} \quad (6)$$

$\bar{g}(t)$ is the complex conjugate of $g(t)$

- Recall (complex conjugate): $z = x + iy \Rightarrow \bar{z} = x - iy$

Fourier series: interpretation

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k \underbrace{\exp\left(\frac{2ik\pi t}{T}\right)}_{\Phi_k(t)} = \sum_{k=-\infty}^{+\infty} c_k \Phi_k(t)$$

- Interpretation of the decomposition into Fourier series:

$$\begin{aligned} c_k &= \frac{1}{T} \int_0^T x(t) \exp\left(\frac{-2i\pi kt}{T}\right) dt \\ &= \langle x, \Phi_k \rangle \end{aligned}$$

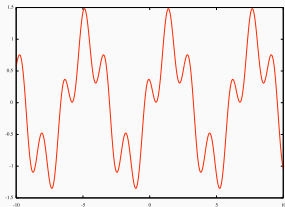
$\Rightarrow c_k$ is the projection of x on the complex function $\exp(\frac{2i\pi kt}{T})$

- Exercise: show that $(\exp(\frac{2i\pi kt}{T}))_{k \in \mathbb{Z}}$ is an orthonormal basis

Fourier series: geometric interpretation

Let x be a signal

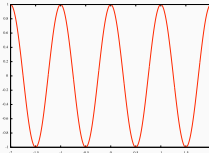
$$\left(x(t) = \sum_{j=1}^4 \cos(2\pi k_j \frac{t}{T})\right)$$



$$x(t) = \sum_k c_k \exp\left(\frac{2ik\pi t}{T}\right)$$

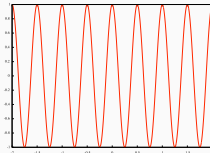
$$c_k = \left\langle x, \exp\left(\frac{2i\pi kt}{T}\right) \right\rangle$$

$$e^{\frac{\pm 2i\pi k_1 t}{T}}$$



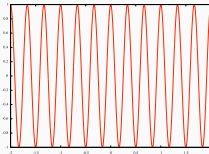
$$c_{k_1} = c_{-k_1} = \frac{1}{2} a_{k_1} = \frac{1}{2}$$

$$e^{\frac{\pm 2i\pi k_2 t}{T}}$$



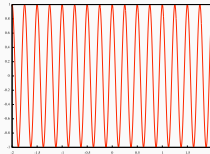
$$c_{k_2} = \frac{1}{2}$$

$$e^{\frac{\pm 2i\pi k_3 t}{T}}$$



$$c_{k_3} = \frac{1}{2}$$

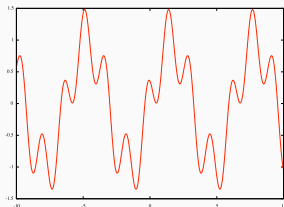
$$e^{\frac{\pm 2i\pi k_4 t}{T}}$$



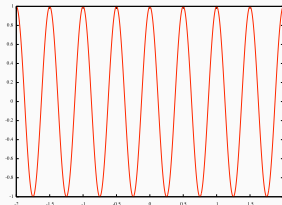
$$c_{k_4} = \frac{1}{2}$$

Fourier series: geometric interpretation

signal $x(t)$



basis function $\Phi_k(t) = \exp(\frac{2ik\pi t}{T})$



$$x(t) = \sum_{k=-\infty}^{+\infty} c_k \exp\left(\frac{2ik\pi t}{T}\right) \iff c_k = \langle x, \Phi_k \rangle$$

$\implies$ the scalar product c_k measures the similarity between the signal x and an element Φ_k of the basis

- c_k is the **amount** of pure frequency $\exp(\frac{2i\pi kt}{T})$ contained in the signal x

Changes the way a function is described

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k \exp\left(\frac{2ik\pi t}{T}\right) \iff c_k = \langle x, \Phi_k \rangle$$

- All periodic functions may be represented in the Fourier basis. A periodic function is then described as an infinite (and countable) set of complex numbers: $c_k, k \in \mathbb{Z}$
- A non null c_k coefficient means: $\exp\left(\frac{2i\pi kt}{T}\right)$ element is contained in signal x with “quantity” c_k
- Example: $x(t) = \cos(2\pi \frac{t}{T}) : c_k = \begin{cases} 1/2 & \text{if } k = \pm 1 \\ 0 & \text{otherwise} \end{cases}$
 - projection of x on $\{\exp(\frac{\pm 2i\pi t}{T})\}$ gives $\frac{1}{2}$
 - projection of x on $\{\exp(\frac{2i\pi kt}{T})\}$ gives $0 \forall k \neq \pm 1$

Context: analysis of 1D signals in the Frequency domain

Fourier transform of signals (1D)

Definition

Mathematical tools and Fourier transform of some usual functions

Fourier transform of images (2D)

2D Fourier transform: application to image processing

Continuous Fourier transform of a signal (1D)

Fourier series $\rightarrow$ Fourier transform

- Any periodic signal can be represented by its Fourier series
- Non periodic signal? limit case of a periodic signal when $T \rightarrow \infty$
- If x is non periodic, the Fourier series does not hold: the countable basis $\{\exp(\frac{2i\pi kt}{T})\}$, $k \in \mathbb{Z}$ must be changed to a continuous basis!
- **Definition:** Fourier transform X of a signal x

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-i2\pi ft} dt \quad \text{with} \quad f \in \mathbb{R} \quad (7)$$

- Definition: X is the frequency response of x
- The Fourier transform is a generalization of Fourier series to non-periodic functions

Continuous Fourier transform of a signal

Interpretation

$$\begin{aligned}X(f) &= \int_{-\infty}^{+\infty} x(t)e^{-i2\pi ft} dt \quad f \in \mathbb{R} \\ &= \langle x, \Phi_f \rangle\end{aligned}$$

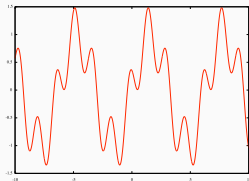
with $\Phi_f(t) = \exp(i2\pi ft)$ and $\langle f, g \rangle = \int_{-\infty}^{+\infty} f(t)\bar{g}(t)dt$

- x is **projected** on a uncountable set¹ of continuous functions $(\exp(i2\pi ft))_{f \in \mathbb{R}}$: f is a real parameter, not a discrete index
- $X(f)$ gives information about the frequency f of signal x :
 - Is the frequency f present in signal x ? To which amount?
 - The variable f is real: a continuum of frequencies versus a discrete set of frequencies for Fourier series

¹a set whose elements cannot be counted, such as $\mathbb{R}$

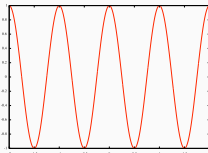
Fourier transform: geometrical interpretation

- A signal $x(t)$



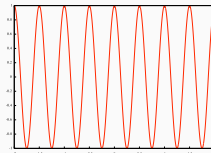
$$X(f) = \langle x, \exp(2i\pi ft) \rangle$$

$$e^{\pm 2i\pi f_1 t}$$



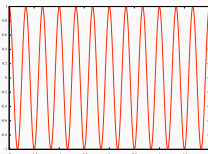
$$X(f_1) = \langle x, \exp(\pm 2i\pi f_1 t) \rangle$$

$$e^{\pm 2i\pi f_2 t}$$



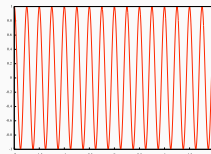
$$X(f_2) = \langle x, \exp(\pm 2i\pi f_2 t) \rangle$$

$$e^{\pm 2i\pi f_3 t}$$



$$X(f_3) = \langle x, \exp(\pm 2i\pi f_3 t) \rangle$$

$$e^{\pm 2i\pi f_4 t}$$

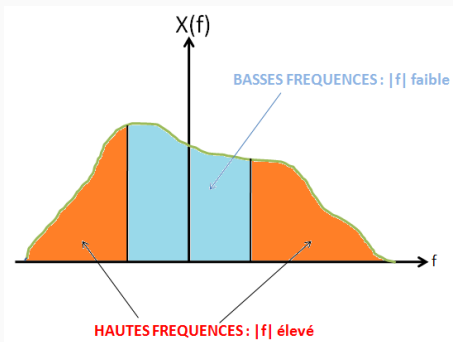


$$X(f_4) = \langle x, \exp(\pm 2i\pi f_4 t) \rangle$$

Fourier transform of a signal

Interpretation: high and low frequencies of a signal

- High frequencies of $x(t)$: significant values of $X(f)$ for high values of $|f|$ means quick variations of $x(t)$
- Low frequencies of $x(t)$: significant values of $X(f)$ for low values of $|f|$ means slow variations of $x(t)$



Fourier transform of a signal

Remarks

- $X(f)$ is a complex number
 - The real part of X is $X_R(f) = \int_{-\infty}^{+\infty} x(t) \cos(2\pi ft) dt$, an even function
 - The imaginary part of X is $X_I(f) = -\int_{-\infty}^{+\infty} x(t) \sin(2\pi ft) dt$, an odd function
- Real and imaginary parts are not interpretable, prefer:
 1. module or amplitude spectrum: $|X(f)| = \sqrt{X_R(f)^2 + X_I(f)^2}$
 $\hookrightarrow$ the norm of $X(f)$ seen as a vector in the complex plane.
 $|X(f)|$ represents the amount of pure frequency f present in signal x
 2. phase: $\Theta(f) = \arctan\left(\frac{X_I(f)}{X_R(f)}\right)$
 $\hookrightarrow$ the angle of $X(f)$ seen as a vector in the complex plane with the real axis. The phase represents the shift between x and $\exp(2i\pi ft)$

Remarks (cont'd)

- A “special” frequency, $f = 0$: $X(0)$ is the mean of the signal!

$$\begin{aligned} X(0) &= \int_{-\infty}^{+\infty} x(t) \exp(-i2\pi 0 t) dt \\ &= \int_{-\infty}^{+\infty} x(t) dt \end{aligned}$$

Signal reconstruction

- The signal x can be reconstructed from its Fourier transform X :

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{i2\pi ft} df$$

- Projection on basis $(\bar{\Phi}_f)_{f \in \mathbb{R}}$, with $\bar{\Phi}_t(f) = \exp(-i2\pi ft)$:

$$x(t) = \langle X, \bar{\Phi}_t \rangle$$

- We can therefore reconstruct the signal by “summing” the projections...
 - Direct consequence of a projection on an orthogonal basis

Context: analysis of 1D signals in the Frequency domain

Fourier transform of signals (1D)

Definition

Mathematical tools and Fourier transform of some usual functions

Fourier transform of images (2D)

2D Fourier transform: application to image processing

Convolution product

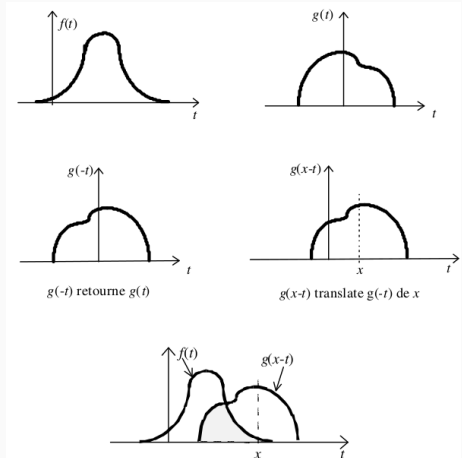
- Operator $\star$ denotes the convolution product taking two signals x and y as operands and returning a signal z :

$$\begin{aligned} z(t) &= x \star y(t) = \int_{-\infty}^{+\infty} x(\tau)y(t - \tau)d\tau \\ z &= x \star y \end{aligned}$$

- Steps:
 1. Signal reversal: $y(\tau) \rightarrow y(-\tau)$
 2. Signal translation by t : $\rightarrow y(t - \tau)$
 3. Product with x : $x(\tau)y(t - \tau)$
 4. Summation of $x(\tau)y(t - \tau)$

Convolution product: geometric interpretation

1. Signal reversal:
 $g(t) \rightarrow g(-t)$
2. Signal translation
by t : $\rightarrow g(x - t)$
3. Product with f :
 $f(t)g(x - t)$
4. Summation of
 $f(t)g(x - t) \rightarrow$
 $f \star g(x)$



Convolution product: important properties

Convolution operator is

- commutative:

$$x \star y = y \star x$$

- distributive:

$$x \star (y + z) = x \star y + x \star z$$

- and associative:

$$(x \star y) \star z = x \star (y \star z)$$

Important properties of 1D Fourier transform

- Linearity: $\text{FT}[t \mapsto ax(t) + by(t)](f) = aX(f) + bY(f)$
- Time scaling: $\text{FT}[t \mapsto x(\alpha t)](f) = \frac{1}{|\alpha|} X\left(\frac{f}{\alpha}\right)$
- Time shifting : $\text{FT}[t \mapsto x(t - t_0)](f) = X(f)e^{-i2\pi t_0 f}$
- Frequency shifting: $\text{FT}[t \mapsto x(t)e^{-i2\pi f_0 t}](f) = X(f - f_0)$
- Convolution theorem (proof in tutorial works):
 - $\text{FT}[t \mapsto x \star y(t)](f) = X(f) \times Y(f)$
 - $\text{FT}[t \mapsto x(t) \times y(t)](f) = X \star Y(f)$ $\Rightarrow$ Very important property showing the equivalence between spatial filtering and filtering in the frequency domain (Lecture 5)
- $x(t) \in \mathbb{R} \Rightarrow$ Hermitian symmetry: $X(f) = \bar{X}(-f)$, implying:
 - $|X(-f)| = |X(f)|$: module of FT (spectrum amplitude) is an even function
 - Phase and imaginary part of FT are odd functions $\Rightarrow$ Signals (and images) are real functions, their amplitude spectrum is always even

Fourier transform of a rectangular function

- Rectangular function (“gate”):

$$\text{Rect}(t) = \begin{cases} 1 & \text{if } |t| \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

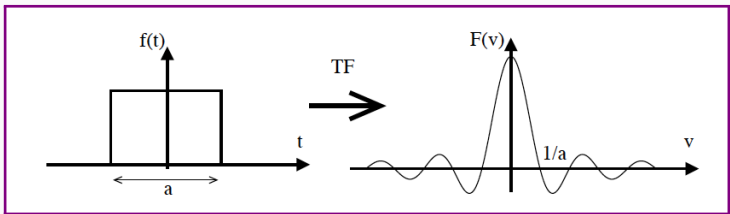
- Fourier transform of a rectangular signal: sinc (sinus cardinal) function (see tutorial works)

$$\text{FT} \left[\text{Rect} \left(\frac{t}{a} \right) \right] (f) = \int_{-\frac{a}{2}}^{+\frac{a}{2}} e^{-i2\pi ft} dt = a \frac{\sin(\pi fa)}{\pi fa} = a \text{sinc}(\pi fa)$$

Fourier transform of some usual signals

Fourier transform of a rectangular function

$$\text{FT} \left[\text{Rect} \left(\frac{t}{a} \right) \right] (f) = a \text{ sinc}(\pi f a)$$



- Very useful for signal digitization – signal windowing and sampling - (Lecture 4)

Fourier transform of a Gaussian function

- Fourier transform of a Gaussian function: also a Gaussian function

$$\text{FT} \left[e^{-b^2 t^2} \right] (f) = \frac{\sqrt{\pi}}{|b|} e^{-\frac{\pi^2 f^2}{b^2}}$$

- The standard deviation in the frequency domain is inversely proportional to the standard deviation in the time domain: a direct consequence of the “time scaling” property

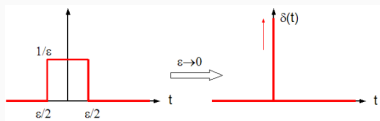
Fourier transform of some usual signals

A very important function: Dirac delta function $\delta(t)$

- Informal (and not rigorous)

definition:

$$\delta(t) = \begin{cases} 0 & \text{if } t \neq 0 \\ +\infty & \text{otherwise} \end{cases}$$

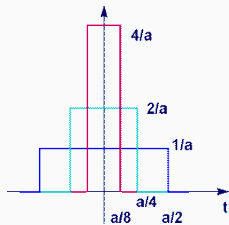


with $\int_{-\infty}^{+\infty} \delta(t) dt = 1$.

More rigorous: $\delta(t)$ is null everywhere except for $t = 0$, has integral equal to 1

- Can be defined as a limit of a sequence of Rectangular functions:

$$\delta(t) = \lim_{a \rightarrow 0} \left[\frac{1}{a} \text{Rect} \left(\frac{t}{a} \right) \right]$$



Dirac function $\delta(t)$ or Impulse

- $\delta(t)$ is not an ordinary function but a **distribution**
 - Generalization of the notion of function
- $\delta(t)$ has a central role in signal and image processing
 - neutral element for convolution product, useful for FT calculus
 - used to model the sampling of signals (1D) and images (2D):

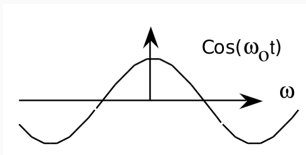
Lecture 4

- Essential properties:
 - $\int_{-\infty}^{+\infty} \delta(t) dt = 1$
 - $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$
 - $x \star \delta(t - t_0) = x(t - t_0)$: translation by t_0 of signal x
 - **Fourier transform:**
 - $\text{FT} [e^{2i\pi f_0 t}] (f) = \delta(f - f_0)$: pure frequency $f_0 \rightarrow$ Dirac
 - $\text{FT} [\delta(t - t_0)] (f) = e^{-2i\pi f t_0}$
 - *Scaling property* : $|\alpha|\delta(\alpha t) = \delta(t)$

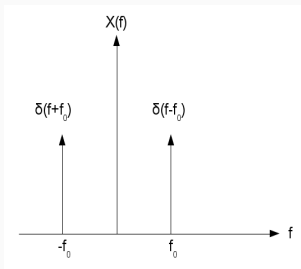
Fourier transform of some usual signals

cos function: $x(t) = \cos(2\pi f_0 t)$

$$\text{FT} [\cos(2\pi f_0 t)] (f) = \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)] \quad (9)$$



$$\omega_0 = 2\pi f_0$$



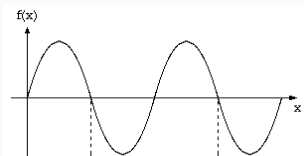
Two pure frequencies $\pm f_0$

- See tutorial works, FT of cos is a real function.

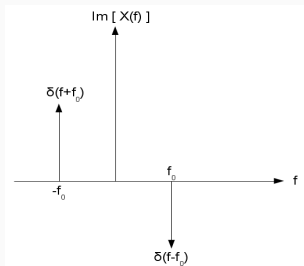
Fourier transform of some usual signals

sin function: $x(t) = \sin(2\pi f_0 t)$

$$\text{FT} [\sin(2\pi f_0 t)] (f) = \frac{i}{2} [\delta(f + f_0) - \delta(f - f_0)] \quad (10)$$



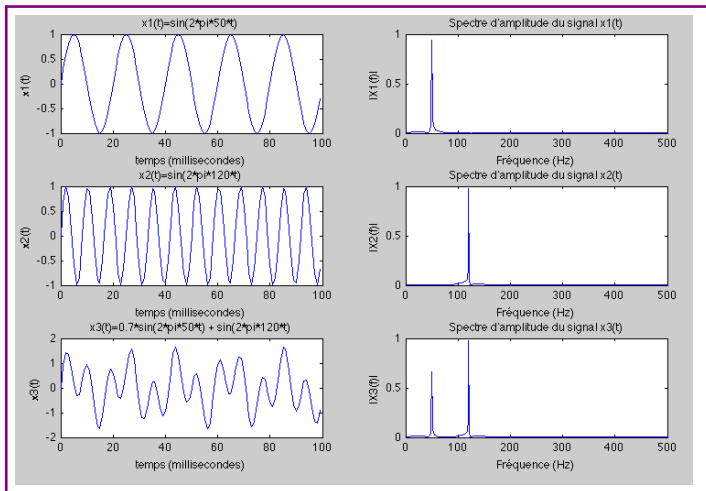
$$x(t) = \sin(2\pi f_0 t)$$



Two pure frequencies $\pm f_0$

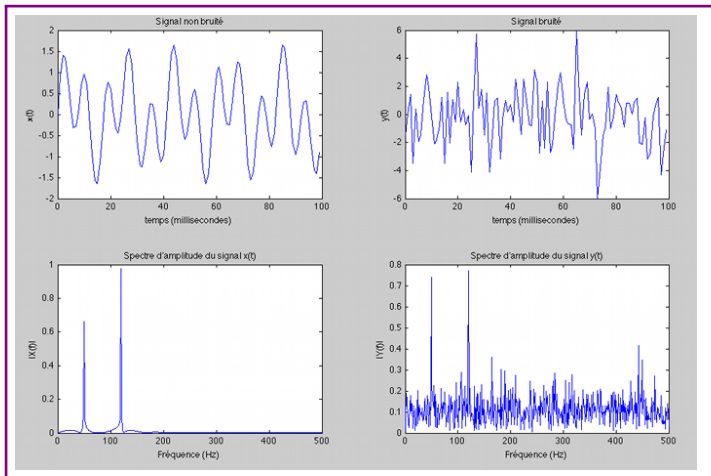
- See tutorial works: FT of sin is an imaginary function

Fourier transform of some usual signals



Fourier transform of some usual signals

An example with noise



Context: analysis of 1D signals in the Frequency domain

Fourier transform of signals (1D)

Fourier transform of images (2D)

Definition

1D vs 2D: similarities and differences

2D Fourier transform: application to image processing

Continuous Fourier transform of a 2D signal

Definition

- Given a 2D and non periodic signal $(t, u) \mapsto x(t, u)$, its Fourier transform is a function $X : \mathbb{R}^2 \rightarrow \mathbb{C}$ defined as:

$$X(f, g) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(t, u) e^{-i2\pi(ft+gu)} dt du$$

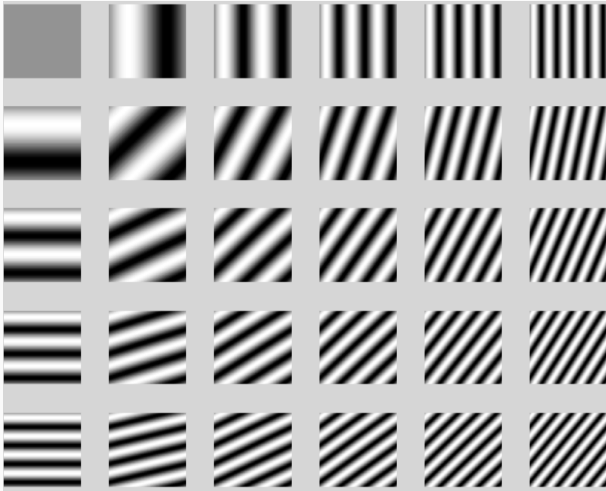
- Interpretation: X is the projection of x on a basis of functions $\{\exp(2i\pi(ft + gu))\}_{(f,g) \in \mathbb{R}^2}$, each function may be viewed as an elementary image:

$$X(f, g) = \langle x, \Phi_{f,g} \rangle$$

with $\Phi_{f,g}(t, u) = e^{i2\pi(ft+gu)}$

Fourier transform of a 2D signal

$\Phi_{f,g}(t, u) = e^{-i2\pi(ft+gu)}$: elementary images



Fourier transform of a 2D signal

Some definitions

- Real part:

$$X_R(f, g) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(t, u) \cos(2\pi(ft + gu)) dt du$$

an even function

- Imaginary part:

$$X_I(f, g) = - \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(t, u) \sin(2\pi(ft + gu)) dt du$$

an odd function

- Module, or amplitude spectrum:

$$|X(f, g)| = \sqrt{X_R(f, g)^2 + X_I(f, g)^2}$$

- Phase: $\Theta(f, g) = \arctan\left(\frac{X_I(f, g)}{X_R(f, g)}\right)$

- The null frequency, $f = g = 0$:

$$X(0, 0) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(t, u) dt du$$

Inverse Fourier transform

- Signal x can be reconstructed from its Fourier representation X :

$$x(t, u) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} X(f, g) e^{i2\pi(ft+gu)} df dg$$

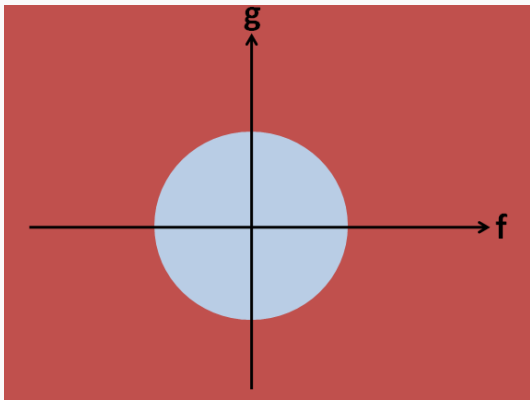
- Direct consequence of a projection on an orthonormal basis:

$$x(t, u) = \langle X(t, u), \bar{\Phi}_{f,g}(t, u) \rangle \quad (11)$$

with $\bar{\Phi}_{f,g}(t, u) = e^{-i2\pi(ft+gu)}$

Fourier transform of a 2D signal

Frequency components

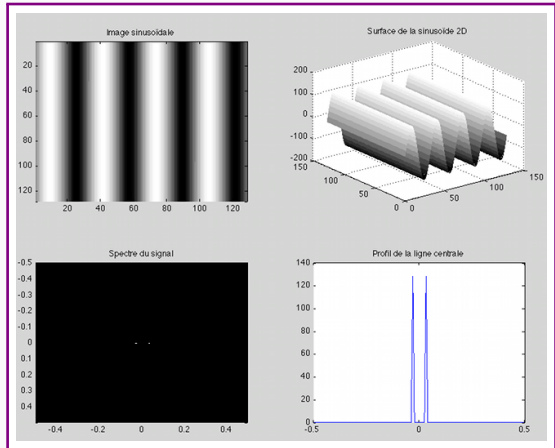


- Low spatial frequency: $f^2 + g^2$ small
- High spatial frequency: $f^2 + g^2$ large

Fourier transform of a 2D signal: first example

$$x(t, u) = \cos(2\pi f_0 t)$$
$$X(f, g) = \frac{\delta(f-f_0) + \delta(f+f_0)}{2}$$

(see tutorial works)

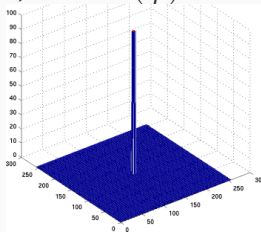
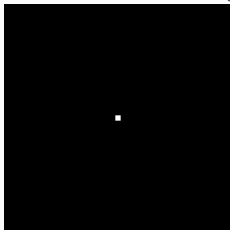


$$\text{FT} [A \cos [2\pi f_o (t \cos(\theta) + u \sin(\theta))]] = ?? \text{ (see tutorial works)}$$

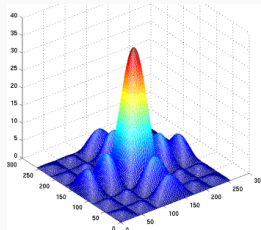
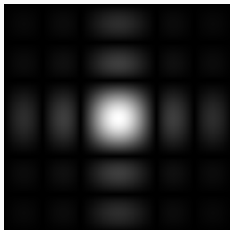
Fourier transform of a 2D signal: second example

2D rectangular function: $x(t, u) = \text{Rect}\left(\frac{t}{T}\right) \times \text{Rect}\left(\frac{u}{T}\right)$

$x(t, u)$



$X(f, g)$



$X(f, g)$ sinc 2D (see tutorial work)

Context: analysis of 1D signals in the Frequency domain

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2D Fourier transform: application to image processing

Fourier transform of a 2D signal

From 1D to 2D

- 2D Fourier transform: a natural extension of the 1D case
 - Many common properties
 - Some 2D specificities
- 2D Fourier transform: $\sim$ 2 successive 1D Fourier transforms (see tutorial works)
 - first transform: $Z(f, u) = \text{FT}[t \mapsto x(t, u)]$
 - second transform: $X(f, g) = \text{FT}[u \mapsto Z(f, u)]$
- if x separable, $x(t, u) = z(t) \times k(u)$, we have:

$$X(f, g) = Z(f) \times K(g)$$

with $Z = \text{TF}[z]$ and $K = \text{TF}[k]$

- Simple product of two mono-dimensional Fourier transforms

1D and 2D FT: common properties

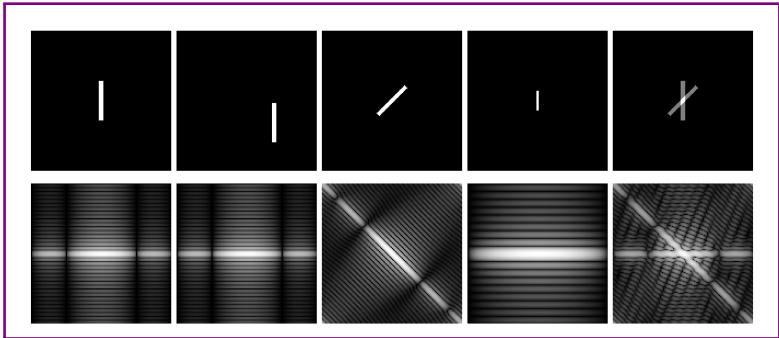
	Continuous 1D FT		Continuous 2D FT	
	$x(t)$	$X(f)$	$x(t,u)$	$X(f,g)$
1	$x(t) + \lambda y(t)$	$X(f) + \lambda Y(f)$	$x(t,u) + \lambda y(t,u)$	$X(f,g) + \lambda Y(f,g)$
2	$x(t - t_0)$	$X(f) e^{-2i\pi f t_0}$	$x(t - t_0, u - u_0)$	$X(f,g) e^{-2i\pi(f t_0 + g u_0)}$
3	$x(\alpha t)$	$\frac{1}{ \alpha } X(\frac{f}{\alpha})$	$x(\alpha t, \beta u)$	$\frac{1}{ \alpha \beta } X(\frac{f}{\alpha}, \frac{g}{\beta})$
4	$x \star y(t)$	$X(f) \times Y(f)$	$x \star y(t,u)$	$X(f,g) \times Y(f,g)$
5	$x(t) \times y(t)$	$X \star Y(f)$	$x(t,u) \times y(t,u)$	$X \star Y(f,g)$

Table 1: Properties of 1D and 2D Fourier transforms

1. linearity
2. translation
3. domain contraction
4. convolution
5. product

1D and 2D FT: common properties

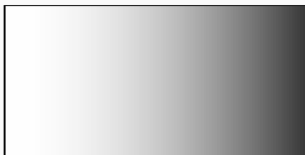
Illustrations



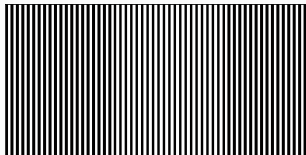
2D Fourier transform: specificities

Notion of “spatial frequency”

- Image $I : \mathbb{R}^2 \rightarrow \mathbb{R}$: function of two variables (spatial position)
- Fourier transform $FT(I) : \mathbb{R}^2 \rightarrow \mathbb{C}$: function of two variables (spatial frequency)
- Spatial frequency $\rightarrow$ couple of frequencies:
 - horizontal frequency: how fast signal varies along x-axis
 - vertical frequency: how fast signal varies along y-axis



Low horizontal frequency
Null vertical frequency

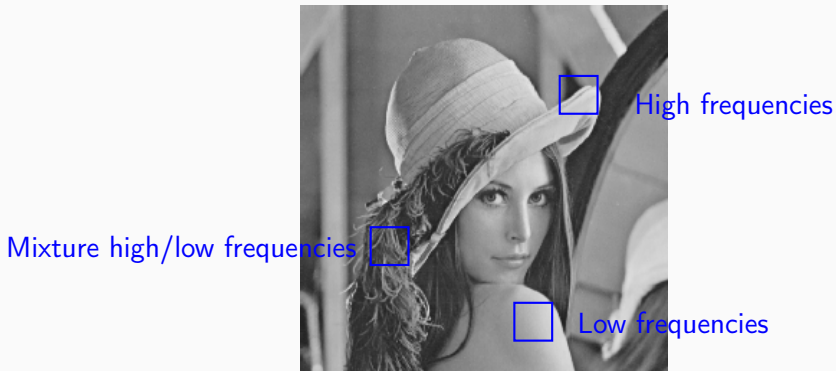


High horizontal frequency
Null vertical frequency

2D Fourier transform: specificities

Notion of “spatial frequency”

- Natural images: non-stationary signals
- Different spatial frequencies (high/low) in different regions of the image



Rotation

- A rotation of angle θ in the spatial domain induces the same rotation in the frequency domain:

$$\begin{aligned}y(t, u) &= x(t \cos \theta + u \sin \theta, -t \sin \theta + u \cos \theta) \\Y(f, g) &= X(f \cos \theta + g \sin \theta, -f \sin \theta + g \cos \theta)\end{aligned}$$

- Important: the frequency response $X(f, g)$ gives an information on the orientation of image structures

2D Fourier transform: example with rotation

barbara



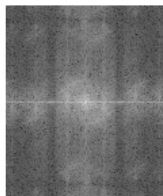
rotation 45°



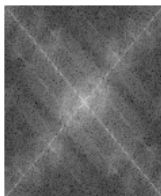
rotation 90°



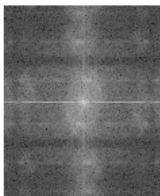
TF 2D



TF 2D

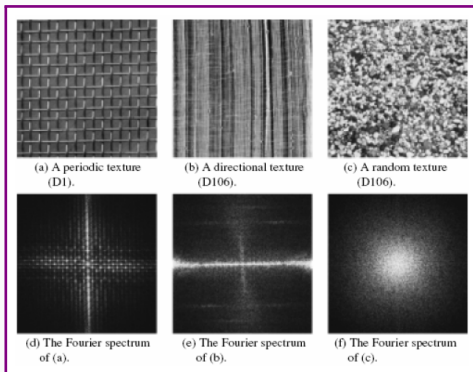


TF 2D



2D Fourier transform: orientations in images

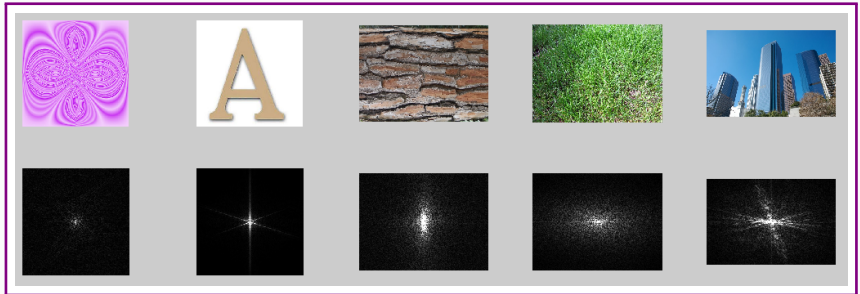
- Important: the frequency response $X(f, g)$ gives an information on the orientation of structures in the image
- Examples with textured images



- Main structure orientations are highlighted in the amplitude spectrum

2D Fourier transform: orientations in images

- Examples with natural images

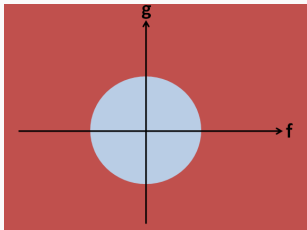


- Main structure orientations are highlighted in the amplitude spectrum

2D Fourier transform: visualizing the spectrum

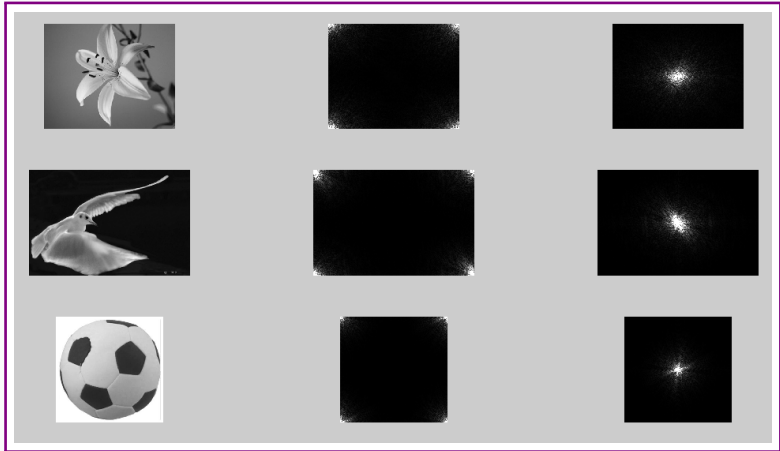
Centering the spectrum:

- Function `fft2()` from `numpy.fft` to compute a 2D Fourier transform
 - `fft2()` is a discrete Fourier transform (DFT) → see Lecture 4
- `fft2()`: spectrum origin at left-top corner of the image
- To shift the origin of the frequencies at the center of the spectrum (more natural visualization), use the function `fftshift()` from `numpy.fft`



- Spectrum values are not changed, only a rearrangement
- Highlights the symmetry property of real signals spectrum ($|F(u, v)| = |F(-u, -v)|$)

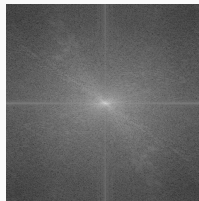
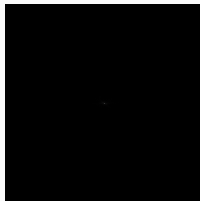
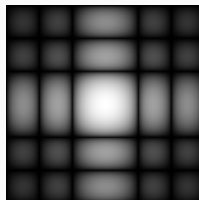
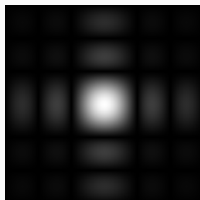
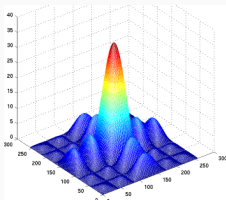
Centering the spectrum: illustration



2D Fourier transform: visualizing the spectrum

Natural images: high/low frequencies

- Energy of low frequencies $\gg$ Energy of high frequencies
 $\Rightarrow$ apply a log transform $\log(|X(f, g)| + 1)$



$$|X(f, g)|$$

$$1 + \log(|X(f, g)|)$$

Context: analysis of 1D signals in the Frequency domain

Fourier transform of signals (1D)

Fourier transform of images (2D)

2D Fourier transform: application to image processing

Fourier transform: application to image processing

- Fourier transform: representation in the frequency domain
- Time representation $\rightarrow$ Frequency representation:
 1. Time (or spatial for 2D case) representation: projection on a basis of Dirac functions $(\Psi_t)_{t \in \mathbb{R}}$, $\Psi_t(u) = \delta(u - t)$:

$$x(t) = x \star \delta(t) = \int_{\mathbb{R}} x(u) \delta(t - u) du = \langle x, \Psi_t \rangle \quad (12)$$

$\Rightarrow x$ is described in time

2. Frequency representation: projection on a sinusoidal basis $(\bar{\Phi}_t)_{t \in \mathbb{R}}$, $\bar{\Phi}_t(f) = e^{-2i\pi ft}$:

$$x(t) = \int_{\mathbb{R}} X(f) e^{2i\pi ft} df = \langle X, \bar{\Phi}_t \rangle \quad (13)$$

$\Rightarrow x$ is described in frequency

- Use the frequency representation if the useful information is better described than with the time representation
- “useful” depends on the context

2D Fourier transform: application to image processing

Noise removal

- Problem formulation: isolate the signal from the noise
- Hypothesis: signal and noise have not the same frequency components, signal $\Leftrightarrow$ low freq, noise $\Leftrightarrow$ high freq
- Method: high frequencies $\leftarrow 0$



Original image



Noisy image



Result image

2D Fourier transform: application to image processing

Compression

- Formulation: represent the signal with fewer data
- Hypothesis: high frequencies have low energy, and contain irrelevant information
- Method: keep only the low frequencies



Original



Compressed image (/2)

⇒ principle of JPEG compression, see practical work 4

2D Fourier transform: application to image processing

Numerous applications of frequency analysis

- Noise removal, compression, image restoration: low level processing
- Useful for higher level processing
- Linear filtering (Lecture 5): convolution in spatial domain $\leftrightarrow$ multiplication in frequency domain: dual operations
- Filtering: preprocessing step for many applications :
 - Edge detection (Lecture 6), detection and description of points of interest, e.g. Harris, SIFT (Lectures 7-8), texture characterization...

Frequency representation: the pro

- well adapted to represent stationary and smooth signals

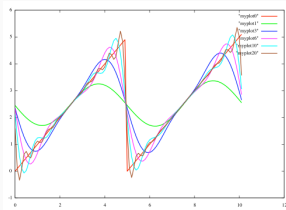
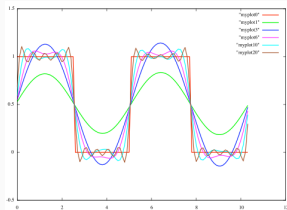
Non stationary signals

- Fourier transform describes the image structures only in terms of frequencies
 - Spectrum amplitude is invariant by translation and cannot localize the image structures in time/space
- Images are non stationary signals: the frequency components are not the same everywhere the image domain (see example on Slide 49)
- Solution: use representations describing the image *both* in time and frequency: wavelets for instance

Frequency representation: the cons

Non derivable signals

- A non derivable signal needs an infinity of frequencies to be described: Gibbs phenomenon may occur if high frequencies are removed



- Not well adapted to describe image contours
- ⇒ JPEG not suited to images having many contours (e.g. text images)
- Solution: a suitable basis to represent edges, again wavelets (JPEG2000)!